QUESTION 1 (10 MARKS)

a) Determine whether the following function is continuous at $x = 0$ or not.

$$f(x) = \begin{cases} \dfrac{x-9}{x^2-3}, & x < 0, \\ \dfrac{\sqrt{x^2+4}}{2}, & x = 0, \\ 3, & x > 0. \end{cases}$$

(3 marks)

b) Without using L'Hopital's rule, evaluate the following limit.

i. $\displaystyle\lim_{x\to 3} \frac{3-\sqrt{x+6}}{\sqrt{x+1}-2}$.

(4 marks)

ii. $\displaystyle\lim_{x\to\infty} \frac{6-x}{\sqrt{x^2+3}+\sqrt{x^2-3}}$.

(3 marks)

QUESTION 2 (20 MARKS)

a) If $y = \operatorname{cosec}^4\left(\dfrac{x^2-x}{e^x}\right)$, find $\dfrac{dy}{dx}$.

(5 marks)

b) If $y = e^{mx}\cos(nx)$ where m and n are constants, show that

$$\frac{d^2y}{dx^2} - 2m\frac{dy}{dx} + \left(m^2+n^2\right)y = 0.$$

(5 marks)

c) Using implicit differentiation, find $\dfrac{dy}{dx}$ in terms of x and y for

$$\ln(xy) + y^2\sin 2x = x^3.$$

(5 marks)

d) By taking natural logarithm on both sides, find $\dfrac{dy}{dx}$ for

$$e^{x+y} = \frac{(x-1)^{\frac{1}{2}}(x+2)^3}{(x+3)^2}.$$

(5 marks)

QUESTION 3 (17 MARKS)

a) The oil drum of cylindrical shape is required to have the capacity to store 0.2 m^3 volume of oil. To manufacture the oil drum, the cost of the material is set depending on the surface area of the parts where RM 5 on the top, RM 5 on the bottom, and RM 10 on the side, per m^2.

i. Show that the cost to produce one oil drum is given by

$$C = 10\pi r^2 + \frac{4}{r},$$

where C is the cost, and r is the radius of the drum.

(3 marks)

ii. Find the radius of the drum that minimize the cost. Hence, find the minimum cost.

(4 marks)

b) A function $f(x)$ is defined as

$$f(x) = 80x + 42x^2 - x^4.$$

i. Find critical points of $f(x)$.

(2 marks)

ii. Classify all the critical points. Hence, sketch the graph of $f(x)$.

(8 marks)

QUESTION 4 (13 MARKS)

a) i. Evaluate $\displaystyle\int \left(\cos^2\theta + \sin\theta\cos^2\theta\right)\, d\theta$.

(3 marks)

ii. Use the substitution $x = 3 + 3\sin\theta$ and result in part (i) to evaluate

$$\int_3^6 \frac{x\sqrt{-(x-3)^2+9}}{27}\, dx.$$

(5 marks)

b) Evaluate $\displaystyle\int \frac{e^x}{\left(e^x+1\right)^2\left(e^x+2\right)}\, dx$ by using the substitution $u = e^x + 1$.

(5 marks)

QUESTION 5 (11 MARKS)

A region R is bounded by four curves $x = \dfrac{y^2}{10} + 2,\ y = -x,\ y = 10$ and x-axis.

a) Sketch the region R.

(3 marks)

b) Find the area of the region R.

(4 marks)

c) Find the volume of the solid generated when the region R is rotated 360° about x-axis.

(4 marks)

QUESTION 6 (12 MARKS)

a) Population of rats in a small city is modelled by the initial value problem

$$\frac{dR}{dt} = 0.07 - 0.1R, \quad R(0) = 7000,$$

where R is the number of rats and t is time in year. Solve the initial value problem. Hence, find the number of rats in the city after 10 years.

(6 marks)

b) Solve the initial value problem

$$\frac{dy}{dx} = \frac{(x^2 \ln x)(y^2 + 1)}{2y}, \quad y(1) = 0.$$

(6 marks)

QUESTION 7 (17 MARKS)

a) Given $f(x) = e^x + 4x^3 - 10x$.

i. By using Intermediate Value Theorem, show that there are roots located in the intervals $(0, 1)$ and $(1, 2)$.

(2 marks)

ii. Show that the iterative formula

$$x_{i+1} = \frac{e^{x_i} + 4x_i^3}{10}, \qquad i = 0, 1, 2, \ldots$$

is suitable to calculate the root with initial guess $x_0 = 0$. Hence, find the root correct to four decimal places.

(5 marks)

iii. Use Newton-Rhapson method to find the second root with initial guess $x_0 = 1$ and $|x_{i+1} - x_i| < 0.0005$.

(5 marks)

b) By using Simpson's rule, compute the integral $\displaystyle\int_0^{\frac{\pi}{2}} \frac{dx}{\sin(x^2) + \cos(x^2) + 1}$ with six subintervals.

(5 marks)

LIST OF FORMULA

Trigonometric	
• $\cos^2 x + \sin^2 x = 1$	• $\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$
• $1 + \tan^2 x = \sec^2 x$	• $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$
• $\cot^2 x + 1 = \operatorname{cosec}^2 x$	• $\tan(x \pm y) = \dfrac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$
• $\sin 2x = 2 \sin x \cos x$	• $2 \sin x \cos y = \sin(x+y) + \sin(x-y)$
• $\cos 2x = \cos^2 x - \sin^2 x$	• $2 \sin x \sin y = -\cos(x+y) + \cos(x-y)$
$= 2\cos^2 x - 1$	• $2 \cos x \cos y = \cos(x+y) + \cos(x-y)$
$= 1 - 2\sin^2 x$	
• $\tan 2x = \dfrac{2 \tan x}{1 - \tan^2 x}$	

Logarithm
$a^x = e^{x \ln a}$
$\log_a x = \dfrac{\log_b x}{\log_b a}$

Differentiations	Integrations
$\dfrac{d}{dx}[k] = 0,\ k$ constant	$\int k\,dx = kx + C,\ k$ constant
$\dfrac{d}{dx}[x^n] = nx^{n-1}$	$\int x^n dx = \dfrac{x^{n+1}}{n+1} + C,\ n \neq -1$
$\dfrac{d}{dx}[e^x] = e^x$	$\int e^x dx = e^x + C$
$\dfrac{d}{dx}[\ln\|x\|] = \dfrac{1}{x}$	$\int \dfrac{dx}{x} = \ln\|x\| + C$
$\dfrac{d}{dx}[\cos x] = -\sin x$	$\int \sin x\,dx = -\cos x + C$
$\dfrac{d}{dx}[\sin x] = \cos x$	$\int \cos x\,dx = \sin x + C$
$\dfrac{d}{dx}[\tan x] = \sec^2 x$	$\int \sec^2 x\,dx = \tan x + C$
$\dfrac{d}{dx}[\cot x] = -\operatorname{cosec}^2 x$	$\int \operatorname{cosec}^2 x\,dx = -\cot x + C$
$\dfrac{d}{dx}[\sec x] = \sec x \tan x$	$\int \sec x \tan x\,dx = \sec x + C$
$\dfrac{d}{dx}[\operatorname{cosec} x] = -\operatorname{cosec} x \cot x$	$\int \operatorname{cosec} x \cot x\,dx = -\operatorname{cosec} x + C$

Parametric Differentiation
$\dfrac{dy}{dx} = \dfrac{dy}{dt} \times \dfrac{dt}{dx}, \qquad \dfrac{d^2y}{dx^2} = \dfrac{d}{dt}\left(\dfrac{dy}{dx}\right) \times \dfrac{dt}{dx}$
Integration by Parts
$\displaystyle\int u\ dv = uv - \int v\ du$
Volume of Cylinder
$V = \pi r^2 h, \quad r = \text{radius}, \; h = \text{height}$
Fixed Point Formula
$x_{i+1} = g(x_i), \quad i = 0, 1, 2, \ldots$
Newton-Raphson Formula
$x_{i+1} = x_i - \dfrac{f(x_i)}{f'(x_i)}, \quad i = 0, 1, 2, \ldots$
Trapezoidal rule
$\displaystyle\int_{x_0}^{x_N} f(x)\,dx = \frac{h}{2}\Big[f_0 + f_N + 2(f_1 + f_2 + \cdots + f_{N-1})\Big],\ h = \frac{x_N - x_0}{N}$
Simpson's rule
$\displaystyle\int_{x_0}^{x_N} f(x)\,dx = \frac{h}{3}\Big[f_0 + f_N + 4(f_1 + f_3 + \cdots + f_{N-1}) + 2(f_2 + f_4 + \cdots + f_{N-2})\Big],\ h = \frac{x_N - x_0}{N}$